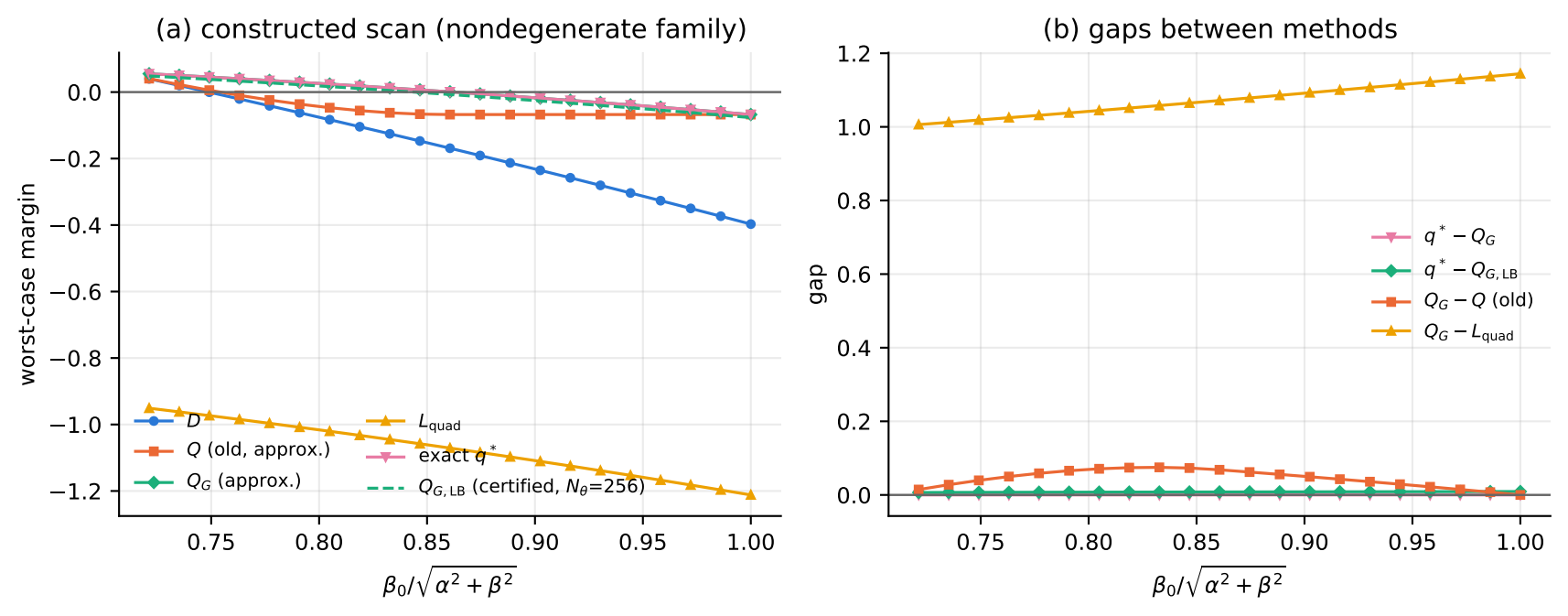
(a) constructed scan (nondegenerate family)
worst-case margin
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
0.75
0.80
0.85
0.90
0.95
1.00
$\beta_0/\sqrt{\alpha^2+\beta^2}$
D
Q (old, approx.)
Q_G (approx.)
L_{quad}
exact q^*
$Q_{G,\mathrm{LB}}$ (certified, N_θ=256)
(b) gaps between methods
gap
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.75
0.80
0.85
0.90
0.95
1.00
$\beta_0/\sqrt{\alpha^2+\beta^2}$
$q^* - Q_G$
$q^* - Q_{G,\mathrm{LB}}$
$Q_G - Q$ (old)
$Q_G - L_{\mathrm{quad}}$